

Mahpara Siddique

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Skills

- **Programming Languages & Frameworks:** Python / PyTorch / TensorFlow / Scikit-Learn / RDKit / Biopython / NumPy / Pandas **AI Architectures & Models:** 3D-EGNN / Transformers (MHSA) / ESM-2 650M / multi-scale CNNs / SVM / Random Forest
- **Bioinformatics & Tools:** PROTAC-DB / PDB / UniProt / RDKit 3D / 3Dmol.js / BLAST
- **Domain Expertise:** Targeted Protein Degradation (PROTACs) / Antimicrobial Peptides (AMP) / Protein Localization

Professional Experience

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|--|-------------------------|------------------------------------|-------------------------------------|
| Teaching Assistant | <u>AI Cadmey</u> | <u>Bahawalpur, Pakistan</u> | (Remote) (11/2024 - 12/2024) |
| <ul style="list-style-type: none">• Taught Fundamental Python programming concepts to students, designing structured coding exercises and interactive quizzes to reinforce core syntax and logic.• Facilitated hands-on programming practice, evaluated student assessments, and provided targeted technical guidance to accelerate coding proficiency. | | | |

Education

Bachelors of Zoology **Bahauddin Zakariya University, Multan, Pakistan** **CGPA- 3.96/4.00** **(10/2019-12/2023)**

Major Coursework: Bioinformatics, Human Genetics, Cell & Molecular Biology, General Biochemistry, Biostatistics

Projects

- **MS-LANet: Multi-Scale Local-Attention Network for Protein Subcellular Localization** ([GitHub Link](#))
Designed MS-LANet to overcome sequence dependency bottlenecks in 10-class protein subcellular localization ($N = 13,907$) by fusing multi-scale 1D CNNs with self-attention mechanisms. Achieved 84.82% accuracy and 0.8120 MCC, outperforming standard baseline models by a +6.87% accuracy margin and $1.21\times$ higher MCC factor.
- **AMP-MHSA: Multi-Head Self-Attention Transformer for Antimicrobial Peptide Discovery** ([GitHub Link](#))
Developed AMP-MHSA to resolve biophysical feature loss in sequence-based antimicrobial peptide discovery ($N = 8,000$) by constructing a Transformer fusing helical positional encodings, Wimley-White transfer energy, and ESM-2 650M embeddings. Reached 93.40% accuracy and 0.8687 MCC, demonstrating an +8.40% accuracy gain and $1.26\times$ higher MCC factor over SOTA baselines.
- **PROTAC-Degrade3D: 3D Equivariant GNN & Ubiquitination Geometry Engine** ([GitHub Link](#)) ([Web deployment Link](#))
Proposed PROTAC-Degrade3D to address 3D spatial ubiquitination transfer bottlenecks in targeted protein degradation ($N = 2,832$) by coupling Satorras 3D-EGNNs with a novel Ubiquitination Geometry Module (UGM). Achieved state-of-the-art 97.46% accuracy and 0.9444 MCC, outperforming Nature Communications baseline DeepPROTACs by +19.51% accuracy ($1.25\times$ MCC factor) and 2D models by +34.25%.

Awards

- Outstanding Performance Award in [AI Agents Hack with LabLab and MindsDB](#) ([GitHub Link](#)) (9/2024)
- Outstanding Performance Award in [Llama Impact Hackathon](#) ([GitHub Link](#)) (12/2024)
- Outstanding Performance Award in [Reasoning with o1](#) ([GitHub Link](#)) (12/2024)
- Outstanding Performance Award in [UC Berkely-Calico coding competition](#) (2024)
- Nobel Service Recognition Award in [Pak Angels' Generative AI Training Course- Cohort6-Mentor](#) (8/2025)
- Nobel Service Recognition Award in [Pak Angels' Generative AI Training Course-Mentor](#) (12/2025)
- Won Harvard University [CS50 Puzzle Day Competition 2026](#) (2026)
- Outstanding Performance Award in [AI Factory Native.builder Hackathon](#) ([GitHub Link](#)) (08/2026)

Certificates

- [Genomic Data Science Specialization](#) (Johns Hopkins University- Coursera) (05/2026)
- [Drug Development Product Management Specialization](#) (University of California San Diego- Coursera) (10/2025)
- [Machine Learning Specialization](#) (DeepLearning.ai -Coursera) (07/2026)
- [Biology Meets Programming](#) (University of California San Diego- Coursera) (11/2025)
- [Generative AI Fundamentals Specialization](#) (IBM- Coursera) (02/2025)

Volunteer Experience

- **Mentor - Pak Angels' Generative AI Training (Remote)** (12/2025)
Taught foundational Python programming to training participants, provided technical mentorship during the capstone Generative AI hackathon, and served as an official judge evaluating final AI project submissions.
- **Moderator -IcodeGuru (Remote)** (2024)
Moderated live online technical sessions, managed audience Q&A, and facilitated smooth execution across multiple community learning cohorts.

Language Skills

- English – Proficient (Reading, Writing, Listening, Speaking)
- Urdu – Native

References

- **Dr Rehana Iqbal**
Associate professor of Zoology, Institute of Zoology
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- **Dr. Fiaz Hussain**
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